

Predicting contaminant transport along quartz veins above the water table in a mica-schist saprolite

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Abstract

Saprolite, thoroughly weathered bedrock, is a porous material that can be used treat wastewater. The inherited veins and fractures in saprolite have been suspected of being capable of transmitting the wastes to ground water. This study evaluated the time of travel of a Br^- solute through a large quartz vein in a mica schist saprolite. Four 150 by 150 cm drainfields were constructed over a saprolite that contained a single quartz vein that was 45 cm wide and continuous across the site. Following saturation of the saprolite, a Br^- tracer and blue-colored dye were applied for a period sufficient for the tracer to leach 30 or 40 cm. Saturated hydraulic conductivity was determined for each drainfield by measuring water intake and also by using the compact constant-head permeameter. The drainfields were then excavated to 90 cm, the dye pattern mapped and soil samples collected for Br^- analysis. Dye patterns below the drainfields indicated that virtually no preferential flow along the vein occurred. Depth of dye penetration within the matrix and vein appeared similar. Deeper penetration did occur along root channels, which extended to 80 cm. There was no significant difference ($P=0.10$) in depth of Br^- penetration in saprolite with and without quartz veins. A simple time of travel model predicted maximum depth of solute movement accurately when no preferential flow occurred. These results showed that large quartz veins that contain clay or Fe–Mn oxides between the quartz gravels will conduct water at a rate that is similar to that found for the saprolite matrix. Such veins pose no apparent hazard to wastewater disposal.

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1. Introduction

Saprolite is a form of weathered bedrock that has been altered to the point that it is porous, can be dug with a spade, and has had a large portion of its

weatherable minerals transformed into to clay (Becker, 1895; Frazier et al., 2002; Kretschmar et al., 1997). Saprolite grades into unweathered bedrock below and soil above. It can form from igneous, metamorphic and sedimentary rocks, and occurs throughout the Piedmont and Mountain regions of the southeastern United States as well as in other warm humid regions worldwide (Cumbie

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and McKay, 1999; Daniels et al., 1999; Eswarin and Bin, 1978; Nahon, 1991; Stolt and Baker, 2000). In terms of its physical and chemical properties, saprolite is a massively structured, permeable material that has chemical and mineralogical properties similar to those found in the overlying soils (Buol et al., 2000).

Waste materials requiring underground disposal or storage are commonly put in saprolite above the water table. As the land available for conventional septic systems decreases, wastewater disposal in saprolite is gaining in popularity in some states. Health regulations will generally not permit its use if the saprolite contains fractures or veins (Amoozegar et al., 1991; Guertal, 1992). Fractures are simple planes of weakness or cracks in saprolite. Veins are tabular or plane-shaped bodies that contain minerals which differ from those in the surrounding saprolite (Guilbert and Park, 1986).

Most saprolites contain geologic fractures and fractured mineral veins that “appear” capable of conducting water rapidly (Pavich, 1986). Colloid-sized particles can move through fractures in saprolite formed from sedimentary rock (Cumbie and McKay, 1999; McKay et al., 2000). Fractures in some saprolites that formed from metamorphic rocks are filled with clay and iron oxides which reduce their saturated hydraulic conductivity to levels found in the saprolite matrix (Buol et al., 2000). The lack of conclusive scientific data on water movement through quartz veins has caused public health officials in North Carolina, for example, to prohibit installation of septic-system drainfields in saprolites that contain veins or fractures. Sites are declared unsuitable for waste disposal if veins or fractures are present within 60 cm (2 ft) of the trench bottom (NCDEHNR, 1999).

Williams et al. (1994) used core samples (7.6 cm in diameter and 7.6 cm in height) to determine hydraulic conductivities of quartz-phyllite saprolite with and without quartz veins. The veins were small, about 1 cm wide and continuous through the cores. There was no statistical difference in the geometric mean hydraulic conductivities between populations. The results suggested that quartz veins were not causing by-pass flow to occur in saprolite, but confirmation could only come from field studies. It was also not known whether the results from this

study applied to longer and wider veins which also occur in saprolite.

Williams and Vepraskas (1994) estimated travel times of solutes moving through the matrix and quartz veins of a quartz-diorite saprolite to determine the relative rates of solute movement between the two zones under field conditions. The veins in this study were up to 12 cm wide and contained gravels which were up to 2 cm in diameter. It was found that veins were not causing preferential flow because solute moved along these veins at a rate similar to that of the saprolite matrix.

Measuring the rate of flow along a vein is difficult but can be accomplished by estimating effective hydraulic conductivities using measurements of the depth of solute movement (Williams and Vepraskas, 1994). The time of travel (TOT) for solutes to move specific distances (L_s) in soil or saprolite by convection or mass flow can be estimated from the average pore water velocity where:

$$TOT = L_s / \bar{v} \quad (1)$$

The average pore water velocity is determined from Darcy's Law as:

$$\bar{v} = [-K(\theta)(\Delta H/L_w)]/\theta \quad (2)$$

where θ is the volumetric water content, $K(\theta)$ is the hydraulic conductivity at the water content θ , and $\Delta H/L_w$ is the hydraulic gradient where ΔH is the difference in total soil water pressure head between two points that are a distance L_w apart. Eqs. (1) and (2) can be combined to determine solute travel times from Darcy's Law as:

$$TOT = -L_s\theta/[K(\theta)\Delta H/L_w] \quad (3)$$

Because $K(\theta)$ represents the average rate of water movement over an area, the above TOT model may perform poorly if $K(\theta)$ is highly variable due to macropores (or quartz veins) conducting solute faster than the soil or saprolite matrix. The TOT model might be improved by using separate K values to account for flow along macropores and matrix. These effective hydraulic conductivity values can be determined experimentally by applying solute to the saprolite for a specific time period, and then observing how deep the solute traveled through the matrix and macropores separately. With such data K_e , the

effective hydraulic conductivity of the macropore or matrix could be calculated by the equation:

$$K_e = -L_s \theta / [T \Delta H / L_w] \quad (4)$$

where L_s is the depth the solute has moved in a vertical profile, T is the time that the solute was applied to the profile, and θ , H and L_w are the same as defined previously.

Evaluation of saprolite for waste disposal includes measuring its saturated hydraulic conductivity (K_{sat}) (Amoozegar et al., 1991). The K_{sat} values can be used to estimate the land area needed for the disposal and treatment of home sewage. Accurate K_{sat} values are needed to optimize land use and ensure that the area used for waste disposal is not too large or too small for a given site. Various methods to measure K_{sat} are available, but it is not always clear whether different techniques provide similar K_{sat} values (Bouma, 1983). Because saprolite occurs below the solum at depths frequently exceeding 2 m, most K data for saprolite have been collected with small cores or in bore-holes of narrow diameter. The concern with using such deep measurements has been that water may be flowing along geologic fractures and veins resulting in by-pass flow.

Previous work on quartz veins had been done on relatively small veins and it was not known whether the results pertained to veins of larger size. The purpose of this study was to determine if large veins (45 cm wide) would conduct water faster than the surrounding saprolite matrix. Specific objectives were: (1) to evaluate water and solute flow along a large quartz vein in a mica-schist saprolite and (2) to evaluate the accuracy of a time of travel model that predicts depth of solute movement under field conditions.

2. Materials and methods

The study was conducted in the North Carolina Piedmont on a mica-schist saprolite that was located in Raleigh on a research farm owned by North Carolina State University (N35°48.298', W78°41.716'). The site was on the sideslope of an upland ridge that was covered by a Cecil clay loam (fine, kaolinitic, thermic Typic Kanhapludults). A profile

description along with physical and chemical data were reported previously by Kretschmar et al. (1997). The soil had A and Bt horizons extending to 71 cm. The Bt horizons were clay loam and silt loam with blocky structures decreasing in grade from strong to weak with depth. The BC and CB horizons spanned depths of 71–310 cm. The BC horizons were silt loam while the CB horizons were sandy loam, with both types having massive rock-controlled structure. Soil pH's in the BC and CB horizons were approximately 4.2, the effective cation exchange capacities were between 3.0 and 5.0 cmol_c kg⁻¹ and base saturations were generally between 10% and 20%. Clay films were observed in the root channels found in these horizons. The C horizon was a loamy sand with massive, rock-controlled structure. Chemical properties were similar to those described for the transitional BC and CB horizons.

Saprolite containing a quartz vein was exposed at the base of a pit that was excavated using a backhoe. The bottom of the pit was carefully cleared and leveled by hand using shovels. Experiments were conducted in small “drainfields” (Fig. 1). Two drainfields were located over a vein, while two others were placed to not include the vein. Because the vein was large (45 cm wide), it was felt the saprolite matrices in plots with veins were different than the matrices away from the veins. Therefore, hydraulic conductivities were measured and compared between plots with veins and plots without veins.

Methods to monitor solute movement along the vein were virtually the same as those used by Williams and Vepraskas (1994) on smaller veins, and much of the following description has been taken from their work. Drainfields were square watertight dikes (150 by 150 cm) made of water-resistant plywood that were assembled and placed over the saprolite of interest (Fig. 1). Dike sidewalls were buried to a depth of 15 cm leaving 15 cm above the surface. The outside edge of the dike was sealed with bentonite pellets. Within the diked drainfield, loose saprolite within 8 cm of the dike walls was compacted to ensure that sidewall leakage would not occur.

Saturated hydraulic conductivity was measured with the compact constant head permeameter at locations shown in Fig. 1 (Amoozegar, 1989). Holes 6 cm in diameter were bored to a depth of 25 cm and filled with water to a depth of 15 cm above the bottom

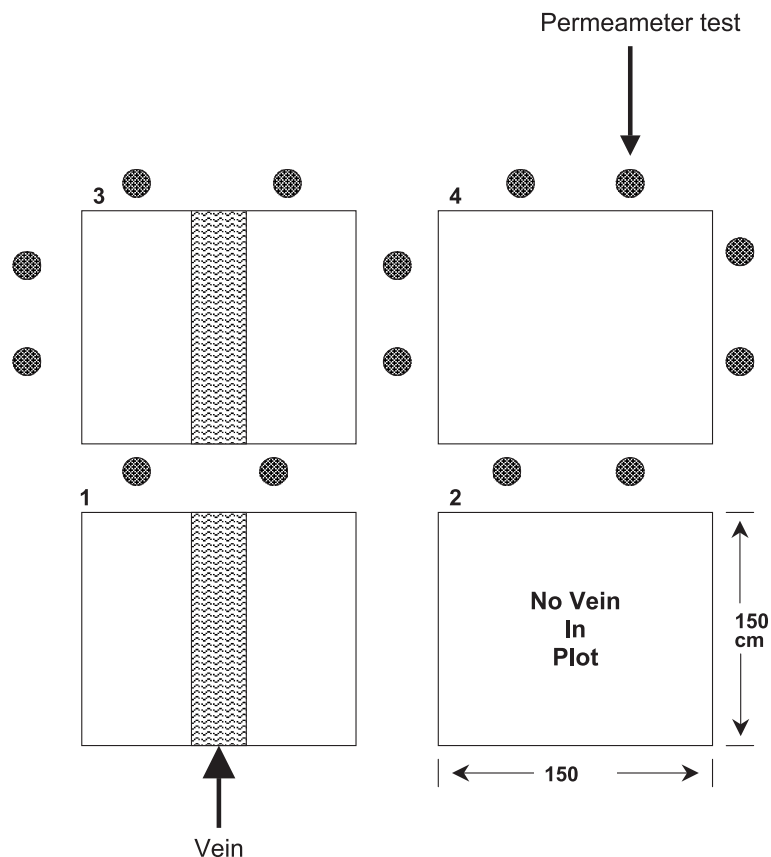


Fig. 1. Plan view of the site in the mica-schist saprolite showing locations of the drainfields with and without veins, along with locations used to measure saturated hydraulic conductivity with the compact constant-head permeameter.

of the holes. The K_{sat} values were estimated at each location using the Glover equation.

Tensiometers were installed in each drainfield to determine hydraulic gradients (Fig. 2). Two tensiometers were installed at each of two depths (30 and 60 cm) near a vein and two were installed away from the vein at the same depths using standard techniques (Cassel and Klute, 1986). Soil water pressure head was measured by pressure transducer (Tensimeter, Soil Measurement Systems, Tucson, AZ). Two neutron probe access tubes were installed to a depth of 60 cm in each drainfield (Fig. 2). The outsides of the tubes were sealed at the drainfield surface with 2 cm of bentonite. A calibration curve was developed outside the drainfields to determine volumetric water content (θ) using standard methods (Gardner, 1986). A layer of coarse sand (5 cm thick) was placed on the instrumented drainfield to protect the surface.

Water application to each drainfield was controlled by a float valve attached to the sidewall of the drainfield and connected by hose to a mobile 1900-l tank. The float valve maintained a 2-cm head of water above the sand-covered surface in the drainfield. When flooded, the drainfield was covered tightly by plastic film to prevent evaporation. After tensiometers at the 30-cm depth indicated that the soil water pressure head was approximately 0 cm and that steady-state conditions had been reached, the volume of water infiltrating into the drainfield over a 24-h period was carefully measured and hydraulic conductivity was determined using the following form of Darcy's equation:

$$K = -Q/[AT\Delta H/L_w] \quad (5)$$

where K is equal to the hydraulic conductivity, Q is the volume of water infiltrating the drainfield, A is the

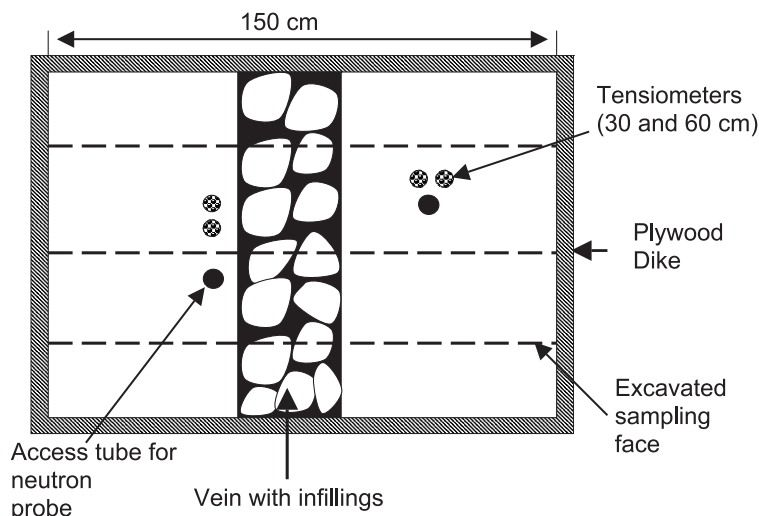


Fig. 2. Plan view of a diked drainfield showing dimensions and instrumentation. Dashed lines represent sampling faces. Tensiometers were installed in duplicate at depths of 30 and 60 cm. Access tubes were installed to 60 cm.

drainfield area, $\Delta H/L_w$ is the hydraulic gradient between the depths of 0 and 30 cm, and T is time.

When K measurements in drainfields were completed, a 0.1% dye solution of Acid Blue 29 was added to the drainfield water to stain water-conducting macropores. After hydraulic gradients stabilized, a 1000 mg l⁻¹ solution of Br⁻ (as KBr) was added to the dye solution in the drainfields for the time determined using Eq. (3) for each drainfield. For Eq. (3), L_s was chosen to be 30 cm for drainfields 1 and 2, but 40 cm for drainfields 3 and 4. For the remaining variables in Eq. (3), $K(\theta)$ was determined as described previously and $\Delta H/L_w$ was the hydraulic gradient between the depths of 0 and 30 cm.

After the solute was applied for a sufficient time period, the remaining surface water was siphoned from the drainfield and all equipment removed. Three vertical faces were exposed with a backhoe in each drainfield to collect soil samples for Br⁻ analysis (Fig. 2). The three sampling faces were approximately 40 cm apart, 90 cm deep and perpendicular to the quartz veins. A sampling grid (120 by 90 cm) with 10 by 10 cm square cells was then placed against the exposed face. Approximately 100 g of moist soil was collected by knife from the center of each grid cell and placed in an air-tight soil moisture can.

Soil samples were oven-dried (105 °C), crushed to pass a 2-mm-sieve, mixed and a 10-g subsample was collected from each for analysis. Fifty milliliters of distilled water were added to each subsample and the suspension was shaken for 15 min. Bromide concentration was determined by a commercial ion analyzer (Orion Research Microprocessor Ionalyzer Model 901, Cambridge, MA) with a Br⁻ electrode that detected Br⁻ in concentrations as low as 0.04 mg l⁻¹. The Br⁻ concentration was expressed as mg l⁻¹ and graphed by sampling grid location for each of the six sampling faces (Fig. 3).

Sampling cells from each profile face were grouped into “miniprofiles” to determine the maximum depths that Br⁻ moved downward through the quartz veins and saprolite matrix. Each miniprofile was a section of the sampling face that consisted of one column of nine sampling cells (10 cm wide) that extended from the drainfield surface to the base of the sampling grid at a depth of 90 cm (Fig. 3). Miniprofile dimensions were 10 by 90 cm. Across each sampling face there were 12 miniprofiles, and each drainfield contained 36 miniprofiles. The maximum depth of Br⁻ penetration was determined separately for each miniprofile. This maximum depth was defined as the lowest cell of a miniprofile that had a Br⁻ concentration ≥ 100 mg l⁻¹ (10% of the 1000 mg Br⁻ l⁻¹ added). This cutoff concen-

	Miniprofile				Miniprofiles with vein				Miniprofile number			
	1	2	3	4	5	6	7	8	9	10	11	12
10	75	26	222	351	<u>271</u>	<u>58</u>	<u>97</u>	<u>30</u>	45	17	156	601
20	170	812	463	539	29	24	4	5	6	16	363	999
30	<u>999</u>	878	273	569	17	4	6	12	6	38	10	346
40	29	<u>501</u>	743	<u>324</u>	24	3	4	<u>139</u>	<u>310</u>	<u>249</u>	434	130
50	4	77	<u>403</u>	25	8	1	1	3	1	2	112	67
60	26	10	16	32	11	13	14	16	17	25	<u>643</u>	954
70	1	2	1	3	1	1	1	1	3	3	80	<u>188</u>
80	1	7	4	1	8	1	1	1	1	1	1	1
90	4	9	16	18		2	2	2	2	2	2	2

Fig. 3. Typical drainfield face with Br^- concentrations (mg l^{-1}) identified in each cell. A miniprofile is represented by the hashed cells. Maximum depth of $100 \text{ mg l}^{-1} \text{ Br}^-$ in each miniprofile is given by the underlined concentration. Miniprofiles 5, 6, 7 and 8 contain a quartz vein which is represented by the double lines (from Williams and Vepraskas, 1994).

tration was selected because Br^- concentrations $< 100 \text{ mg l}^{-1}$ were close to background Br^- concentrations in some drainfield faces. The Br^- concentrations in each miniprofile were examined in order from the bottom-most cell to the top-most cell to identify the maximum depths. Fig. 3 illustrates maximum depths of Br^- penetration for 12 miniprofiles that were defined using this procedure. To evaluate the accuracy of Eq. (3), the maximum depths of Br^- penetration found for each miniprofile were compared to target values. Because Br^- samples were collected from a square 10 by 10 cm grid cell, the samples collected from what we term “the 40-cm depth” represented Br^- that was actually collected between the depths of 30–40 cm.

An effective hydraulic conductivity, K_e , was computed for each miniprofile with Eq. (4). The variable L_s was equal to the observed maximum depth of Br^- penetration found for that miniprofile, θ was the water content determined by neutron probe, T was the time interval over which the Br^- was applied, and $\Delta H/L_w$ was computed for depths of 0 and 30 cm. The maximum depth of Br^- penetration and K_e values were compared among the miniprofiles using analysis of variance procedures (SAS Institute, 1985). Each

drainfield was considered to be one experimental replicate.

3. Results and discussion

The portion of the mica-schist saprolite evaluated occurred between 1.6 and 2.6 m below the ground surface. The quartz vein was up to 45 cm wide and extended laterally for more than 9 m. Quartz fragments were up to 25 cm in diameter. Like the veins evaluated by Williams and Vepraskas (1994), the gravels were separated by infillings of clay and Fe/Mn oxides. Presence of Mn oxides was confirmed in the field by observing a strong reaction to applied H_2O_2 (Jackson, 1956).

Hydraulic data from the drainfield experiments (Table 1) showed that two of the drainfields (numbers 1 and 2) were unsaturated and had soil water pressure heads of -37 and -38 cm at a depth of 30 cm. During excavation the upper 10 cm of the saprolite in these two drainfields was found to contain more clay infillings in root channels and other macropores than occurred in the deeper portion of the saprolite. This infilled zone probably had a lower K_{sat} than the

Table 1

Values for water content, soil water pressure head, hydraulic gradient and hydraulic conductivity for the four drainfields used in the mica-schist saprolite

Drainfield	Vein present	Drainfield measurements				Permeameter hydraulic conductivity	
		Water content ($\text{cm}^3 \text{cm}^{-3}$)	Soil water pressure head ^a (cm)	Hydraulic gradient (cm cm^{-1})	Hydraulic conductivity (cm h^{-1})	Mean (cm h^{-1})	Minimum to maximum (cm h^{-1})
1	Yes	0.42	−37	−2.2	0.6	—	—
2	No	0.49	−38	−2.1	0.4	—	—
3	Yes	0.41	+2	−1.3	0.6	0.8	0.1–2.7
4	No	0.53	−8	−1.6	0.3	0.4	0.3–0.8

^a Measured at a depth of 30 cm.

material below and kept the lower saprolite from saturating. The other two drainfields were placed deeper into saprolite to avoid the infilled zone. Their soil water pressure heads were positive or slightly negative, and had hydraulic gradients closer to -1 cm cm^{-1} . The hydraulic conductivities of the drainfields with and without veins were similar and were approximately 0.5 cm h^{-1} despite the differences in soil water pressure heads. Drainfield K measurements were very similar to those measurements with the permeameter despite the latter using a much smaller volume.

Dye was used to stain water-conducting pores during the time water was applied to the drainfields.

During excavation of the drainfields, grid cells containing dyed soil were identified and mapped (Fig. 4). The dye pattern near the surface showed that most soil particles were stained blue because of water flow through virtually all pores in the matrix. The maps showed the dye moved downward in approximately a uniform front. There was no apparent movement along fractures or any feature related to rock structure. Fingers of dyed material were seen up to 10 cm below the dyed front. Location of the fingers appeared to be random. They were not related to rock structure and no obvious feature was producing preferential flow where the tongues occurred. The depth of staining along the vein generally matched the

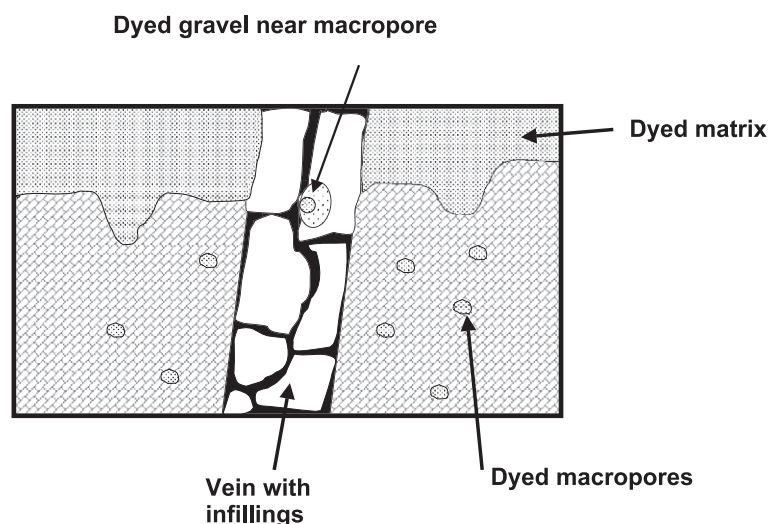


Fig. 4. Diagram showing the dye pattern observed in vertical exposure of saprolite containing a quartz vein. The depth of the dye front in the matrix and vein was similar across the exposure. Tongues of dyed material extended up 10 cm beyond the main front. Dyed macropores (root channels) were found at depths up to 80 cm from the drainfield surface. Gravels in vein were stained by dye if they were near macropores.

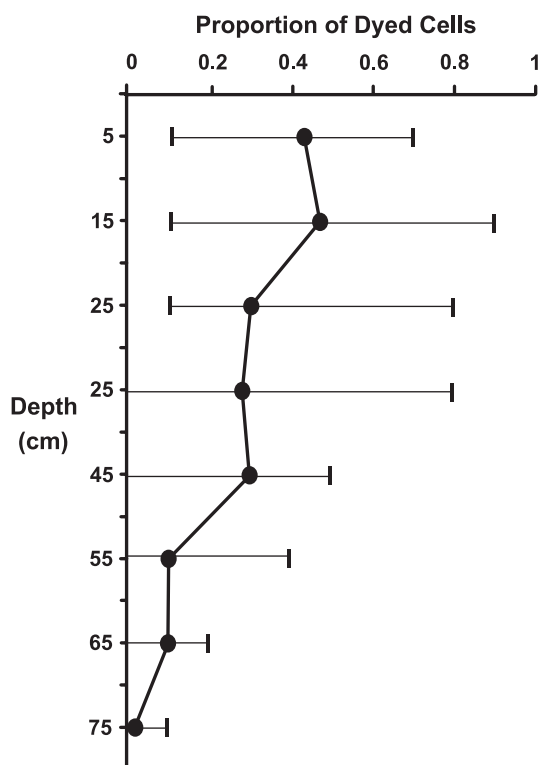


Fig. 5. Relative abundance of water-conducting channels (≥ 1 mm diameter) with depth. Data were obtained from both drainfields by putting a grid of 108 cells (9 rows of 12 cells) on 6 vertical exposures and observing whether each cell contained dye-stained channels. Mean and standard deviations are shown.

depth of staining in the matrix. This showed that the vein did not impede flow nor was water flowing preferentially along the vein while bypassing the matrix. Below the dyed front the macropores were stained to depths of 80 cm. The macropores consisted of root channels with diameters >1 mm. No stained macropores were observed in either drainfields 1 and 2, probably because the drainfields were unsaturated and the macropores did not fill with water. Dyed macropores were found throughout drainfields 3 and 4 and showed no apparent preference for the vein. The dye patterns showed that some movement in the veins did occur for distances <30 cm below the dye front. These occurred where there were spaces between quartz gravels. It was not clear why such spaces were found within the vein but it is possible that settling during weathering may occasionally open gaps within the vein. It is also possible that some gaps

between gravels were created during the initial grading and exposure of the vein at the site. The effects of these gaps on water movement apparently are insignificant because they are essentially isolated voids.

The means and ranges of stained macropores for specific depths that were found in drainfields 3 and 4 are shown in Fig. 5. The root channels were found in all excavated faces in the drainfields. The distribution shown in Fig. 5 is virtually identical to that found in a quartz-diorite saprolite in a previous study (Vepraskas and Williams, 1995).

Results from the Br^- application experiments in the saprolite are reported in Table 2 and Fig. 6. Two different target depths (30 and 40 cm) were used for Br^- penetration and the experiments for each are referred to as “test 1” and “test 2” (Table 2). The distributions of Br^- penetration appeared to be log-normally distributed (Fig. 6), and so logs of the maximum Br^- penetration depths were used in statistical comparisons, and geometric means are reported in Table 2. The geometric mean for maximum depth in Br^- penetration was significantly less in the plot containing the vein than the plot without the vein, although the effective hydraulic conductivities of the two plots were not significantly different. The Br^- was applied for about one-half the time in the plot with the vein than was used for the

Table 2

Means for maximum depths of Br^- penetration, duration of Br^- application and effective hydraulic conductivities in the four drainfields examined in the mica-schist saprolite

Drainfield number	Vein present	Mean depth of maximum Br^- penetration (cm)	Duration of Br^- application (h)	Effective hydraulic conductivity (cm h^{-1})
<i>Test 1</i>				
1	Yes	17	9	0.4
2	No	26	17	0.4
	Difference	*		NS
<i>Test 2</i>				
3	Yes	36	21	0.5
4	No	35	43	0.3
	Difference	NS		NS

* Means were significantly different at the $P=0.05$ level as determined by a two-tailed t -test. NS indicates means were not significantly different at the $P=0.10$ level.

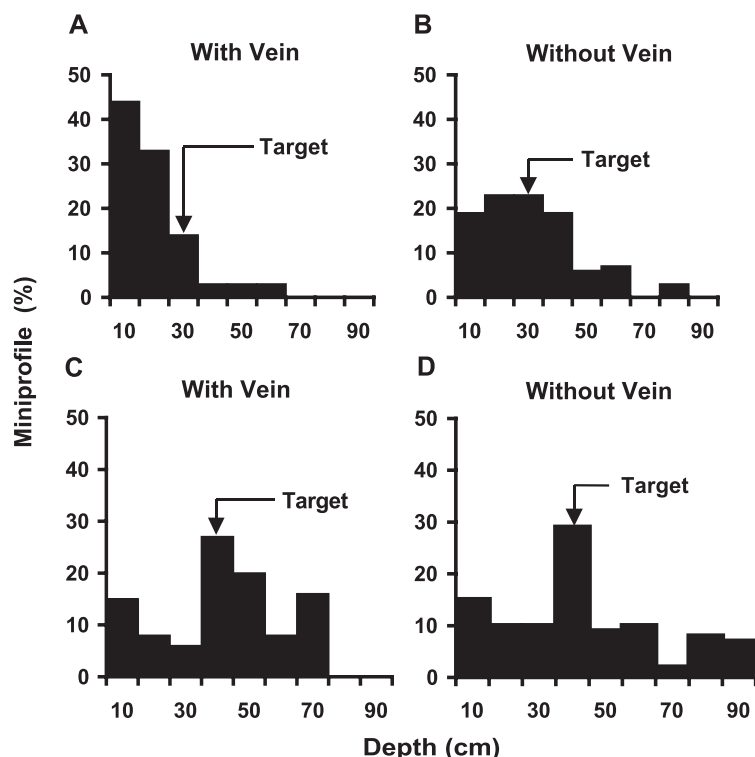


Fig. 6. Frequency distributions of maximum depth of Br^- penetration in miniprofiles of the mica-schist saprolite for: (A) drainfield 1 with a vein, (B) drainfield 2 without a vein, (C) drainfield 3 with a vein and (D) drainfield 4 without a vein.

plot without the vein. The time difference accounted for the Br^- moving a shorter distance where the vein was present. The short application period was determined by using the measured hydraulic conductivity of 0.6 cm h^{-1} (Table 1). This value was 50% greater than the effective K of 0.4 cm h^{-1} and caused us to apply Br^- for a shorter time to the plot with the vein. In test 2, the depth of Br^- penetration was similar between the two plots and there was no significant difference in K_e between the two plots.

Solute was applied for a duration sufficient for it to travel to a depth of 30 or 40 cm as predicted by Eq. (3). The model's prediction was considered successful if the observed maximum depth of Br^- penetration was ≤ 40 cm when our target depth was 30 cm and was ≤ 50 cm for a target depth of 40 cm. Failure occurred if the maximum depth of Br^- penetration exceeded these limits, because for waste disposal purposes adequate purification of effluent does not occur if it travels too quickly. These ranges were selected arbitrarily but account for the occur-

rence of fingering and retarded solute movement due to the wetting front encountering horizontal clay layers, which impede flow.

The success of the TOT model is reported in Table 3. The proportion of miniprofiles that had maximum Br^- penetration depths within the target range were >0.70 for both tests. The time of travel estimates were in general similar for the predicted values and effective values. The largest difference in relative terms occurred in test 1 for the plot with a vein, and as noted earlier the difference is due to the discrepancy between the measured and effective K values.

Using the depth of solute penetration in the miniprofiles, effective hydraulic conductivities were estimated for saprolite containing veins and macropores (Table 4). The K_e values for both drainfields containing veins showed that the vein and matrix conducted water at similar rates, which were not significantly different (0.10 level). Saprolite containing macropores had the same mean K_e as that of the vein.

Table 3

Success of the time of travel model on predicting Br^- penetration depth for two target depths, along with a comparison of predicted and effective time of travel estimates for the four drainfields of the mica-schist saprolite

Drainfield number	Vein present	Target range (cm)	Miniprofiles with Br within target range (Proportion of profiles)	Time of travel (h)	
				Predicted	Effective
<i>Test 1</i>					
1	Yes	≤40	0.94	10	14
2	No	≤40	0.85	18	18
<i>Test 2</i>					
3	Yes	≤50	0.75	21	25
4	No	≤50	0.72	44	44

Predicted travel times were computed using the data in Table 1, while effective travel times were computed using K_e values rather than the measured K values shown in Table 1.

The results of this study have shown that quartz veins that were 45 cm in width, and contained finer material between the quartz gravels, conducted water at a rate comparable to the saprolite matrix. These veins posed no apparent hazard to on-site wastewater disposal. The Fe/Mn oxides and clay observed between the gravels effectively slow water movement through the vein. These results should be applicable to all veins and fractures that contain such infillings. It is not known whether veins and fractures that are deeper in the saprolite would conduct water and solutes at rates comparable to those of the saprolite matrix. It is possible that deeper veins would have fewer and thinner coatings of clay and therefore would conduct water and solutes at rates greater than those measured here. However, the materials filling the veins and fractures were present at a depth of 4.5 m below the

surface, they appeared fresh, and probably extended much deeper into the saprolite.

These results are similar to those of earlier studies on saprolites that contained smaller veins and fractures (Schoeneberger and Amoozegar, 1990; Vepraskas et al., 1991; Williams et al., 1994; Williams and Vepraskas, 1994). All of these studies examined water flow patterns through saprolites weathered from crystalline rock that are found beneath soils of the Piedmont region in the south-eastern U.S. The saprolite has been intensively weathered such that the saprolite matrix is friable and porous, and the veins and fractures contain infillings of finer material. In such materials the veins and fractures do not create preferential pathways for water movement. On the other hand, studies of saprolite formed from sedimentary rocks show that fractures do conduct water, probably because the fractures are not infilled with finer materials (Cumbie and McKay, 1999). In the western U.S., where weathering has not been so intense, vein plugging is rare and channeling within fractures does occur (Frazier et al., 2002; Graham et al., 1997).

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Table 4

Comparison of effective hydraulic conductivities (K_e) of selected features in two drainfields that contained quartz veins

Drainfield	Comparison 1			Comparison 2		
	Vein (cm h ⁻¹)	Matrix (cm h ⁻¹)	Difference (cm h ⁻¹)	Matrix with macropores (cm h ⁻¹)	Matrix without macropores (cm h ⁻¹)	Difference (cm h ⁻¹)
1 ^a	0.3	0.5	NS	—	—	—
4	0.7	0.6	NS	0.7	0.6	NS

Macropores were not stained in drainfield 1 because the material was unsaturated and macropores apparently did not conduct water. No significant differences were found in K_e values at the 0.10 level.

^a Drainfield 1 was unsaturated while drainfield 4 was saturated.

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